

VISVESVARAYA TECHNOLOGICAL UNIVERSITY - BELAGAVI

S. S. Education Trust's

S.G Balekundri Institute of Technology

Shivabasava Nagar Belagavi, Karnataka, India – 590010



Department of Computer Science and Business System

Final Internship Review

PROJECT TITLE

Enterprise Resource Planning Portal

Internship carried out at
"Vrishanksoft (OPC) PVT LTD"

Student Name:

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Outline

- Company profile
- Introduction
- Methodology
- ERP Portal Features
- Technology stack
- Future Scope
- Conclusion
- References

Company profile

Title: Company Overview – Vrishanksoft (OPC) Pvt. Ltd.

- Vrishanksoft (OPC) Pvt. Ltd. is a software development company registered under the Companies Act, 2013 and listed on the MCA portal, Government of India
- Headquartered in Chikodi, Karnataka, India
- Focuses on delivering innovative and scalable software solutions

Core Service Areas:

- Software Development
- Android Application Development
- Web Development
- Embedded Software Solutions

- Key Strengths:
 - Agile and lean startup-based development approach
 - Focus on rapid MVP delivery
 - Strong emphasis on client satisfaction and feedback-driven development
- Team Structure:
 - Total Team Size: 10 members
 - Includes developers, senior engineers, and client relationship management
- Vision:
 - To become a leading software solution provider
 - Deliver reliable, scalable, and high-quality systems
 - Build long-term trust with clients and stakeholders
- Mission:
 - Deliver Minimum Viable Products (MVPs) rapidly
 - Adopt lean development methodologies
 - Continuously improve through real-time client feedback

INTRODUCTION

- The ERP Portal is a centralized academic management system developed for educational institutions.
- It streamlines key academic processes such as:
 - Attendance tracking
 - Internal marks management
 - Notes and resource sharing
- Provides role-based access for:
 - Teachers
 - Head of Department (HOD)
 - Students
 - Ensures efficient data handling, reduced manual work, and improved accessibility

Problem Statement

- Educational institutions often rely on manual or semi-digital systems for managing academic data
- Data like attendance, internal marks, and notes are scattered across multiple platforms
- Lack of a centralized system leads to:
 - Data inconsistency
 - Increased manual workload for faculty
 - Difficulty in tracking student performance
- Lack of proper **data security and access control** in traditional systems, leading to risks like unauthorized data access, data loss, or misuse of sensitive academic information
- There is a need for a secure, scalable, and centralized ERP solution

ERP Portal Features

- Role-Based Access Control:
- Teachers:
 - Manage attendance, notes, and internal marks
- HOD:
 - Approve teachers
 - Add subjects (papers)
 - Full administrative control
- Students:
 - View notes, attendance, and marks
 - Enroll or withdraw from subjects

- Academic Management:
 - Centralized system for:
 - Attendance records
 - Internal marks
 - Academic resources
- Secure Authentication:
 - Role-based login system
 - Protected routes and user validation

- PDF Export System:
 - Generate and download internal marks reports
- Advanced Student Search:
 - Multi-parameter search functionality
 - Real-time filtering and results
- Bulk Operations:
 - Batch update of student records
 - Efficient handling of large datasets

Student Login

Student

Faculty

Admin

Student Email

Password

[Forgot Password?](#)

Login →]

Home

Student

Faculty

Branch

Notice

Exam

Subjects

Admin



Sundar Pichai

Employee ID: 123456

System Administrator (Super Admin)

[Update Password](#)

Personal Information

Email
admin@gmail.com

Phone
1234567890

Gender
Male

Blood Group
O+

Date of Birth
January 1, 1990

Joining Date
March 19, 2026

Salary
₹50,000

Status
Active

Role
Super Admin

Address Information

Address
123 College Street

City
College City

State
State

Pincode
123456

Country
India

Emergency Contact

Name
Emergency Contact

Relationship
Spouse

Phone
9876543210

Home

Student

Faculty

Branch

Notice

Exam

Subjects

Admin

Student Management



Enrollment Number

Enter enrollment number

Name

Enter student name

Semester

Select Semester

Branch

Select Branch

Search



Please select at least one filter to search students

Technology Stack

- Frontend:
 - React.js
 - Next.js
 - Tailwind CSS
- Backend:
 - Node.js
 - Express.js
- Database:
 - MongoDB
- Security & Tools:
 - JWT Authentication
 - bcrypt (password hashing)
 - Cloudinary (media management)

Future Scope

- Integration with **AI/ML** for **student performance prediction**
- Mobile application for **better accessibility**
- Advanced analytics dashboard for:
 - Faculty insights
 - Student progress tracking
- Implementation of:
 - Notifications (email/SMS)
 - Real-time alerts
- Cloud deployment for **high scalability and availability**

Conclusion

- The ERP Portal successfully provides a **centralized academic management system**
- It simplifies key processes like:
 - Attendance tracking
 - Internal marks management
 - Resource sharing
- Role-based access ensures **security and controlled data access**
- The system reduces manual effort and improves **efficiency and accuracy**
- Built using modern technologies, making it **scalable and user-friendly**

References

- React – <https://react.dev/>
- Next.js – <https://nextjs.org/docs>
- Node.js – <https://nodejs.org/en/docs>
- Express.js – <https://expressjs.com/>
- MongoDB – <https://www.mongodb.com/docs/>
- Authentication & Security:
 - JSON Web Token – <https://jwt.io/introduction>
 - bcrypt – <https://www.npmjs.com/package/bcrypt>
- Development Resources:
 - Stack Overflow – <https://stackoverflow.com/>
 - GitHub Repositories & Technical Blogs

THANK YOU